

Supplementary Information for Emergent cohesion via self-caging in maximally entangled rod packings

Yeonsu Jung¹ and L. Mahadevan^{1,2,3}

¹ School of Engineering and Applied Sciences,
²Department of Physics, ³Department of Organismic and Evolutionary Biology,
Harvard University, Cambridge, MA, USA

S1 Captions for Supplementary Videos

1. The evolution of configuration during entanglement maximization followed by relaxation ($N = 100$, $\alpha = 100$).
2. Emergent cohesion via self-caging ($N = 200$, $\alpha = 300$, $N/(Z\alpha) = 0.17 < 1/3$, $Z = 4$). The packing is perturbed under Eq. 12 with $\mu = 0.4$. The translational and rotational gap-to-lengths (Eq. 9) are 0.010 and 0.043, respectively. The reduced gaps (g/μ) are 0.025 and 0.11, indicating rapid dissipation of relative velocities on the sliding timescale. The normalized entanglement is $\tilde{e} = 0.43$.
3. Failure in fragile packings ($N = 200$, $\alpha = 25$, $N/(Z\alpha) = 2 > 1/3$, $Z = 4$). The packing is perturbed under Eq. 12 with $\mu = 1.0$. The translational and rotational gap-to-lengths are ∞ and 0.94, respectively, yielding reduced gaps of ∞ and 0.94. Relative velocities dissipate slowly, as finite gaps do not persist for escaping rods. The normalized entanglement is $\tilde{e} = 0.10$.
4. Failure in woven packings ($N = 331$, $\alpha = 1000$, $N/(Z\alpha) = 0.083$, $Z = 4$). The packing is perturbed under Eq. 12 with $\mu = 1.0$. The translational and rotational gap-to-lengths are ∞ and 0, respectively, with reduced gaps ∞ and 0. Dissipation is limited due to unconfined boundary rods. The normalized entanglement is $\tilde{e} = 0.33$. A small number of boundary rods with $\tilde{g}_t = \infty$ govern the fragility, while interior rods have $\tilde{g} = 0$ and are effectively immobilized.

S2 Rod configuration, contact geometry, and dynamics

A rod i is modeled as a capped cylinder of diameter d and length l . Its centerline is

$$\mathbf{r}_i(s) = \mathbf{c}_i + s \mathbf{t}_i, \quad s \in [-l/2, l/2], \quad (\text{S1})$$

where $\mathbf{c}_i = (x_i, y_i, z_i)$ is the centroid, and $\mathbf{t}_i = (\sin \theta_i \cos \phi_i, \sin \theta_i \sin \phi_i, \cos \theta_i)$ is the unit direction specified by polar angle θ_i and azimuthal angle ϕ_i . The generalized coordinate of rod i is $\mathbf{q}_i = (x_i, y_i, z_i, \theta_i, \phi_i)$, and the full packing state is

$$\mathbf{q} = \begin{bmatrix} \mathbf{q}_0 \\ \mathbf{q}_1 \\ \vdots \\ \mathbf{q}_{N-1} \end{bmatrix} \in \mathbb{R}^{N \times 5}. \quad (\text{S2})$$

The generalized velocity of rod i is the twist

$$\mathbf{v}_i = \begin{bmatrix} \dot{\mathbf{c}}_i \\ \boldsymbol{\omega}_i \end{bmatrix}, \quad (\text{S3})$$

where $\boldsymbol{\omega}_i$ is the angular velocity, so $\dot{\mathbf{t}}_i = \boldsymbol{\omega}_i \times \mathbf{t}_i$ and the velocity of a material point is $\dot{\mathbf{r}}_i(s) = \dot{\mathbf{c}}_i + \boldsymbol{\omega}_i \times (s \mathbf{t}_i)$.

The minimum distance between the centerlines of rods i and j is

$$d_{ij} = \min_{\substack{s \in [-l/2, l/2] \\ t \in [-l/2, l/2]}} \|\mathbf{r}_i(s) - \mathbf{r}_j(t)\|, \quad (\text{S4})$$

and the signed distance function is defined as

$$g_{ij}(\mathbf{q}) = d_{ij}(\mathbf{q}) - d, \quad (\text{S5})$$

with the admissible configuration set given by $g_{ij} \geq 0$ for all $i < j$.

For skew rods ($\mathbf{t}_i \times \mathbf{t}_j \neq \mathbf{0}$) whose closest points lie in the rod interiors ($s^* \in (-l/2, l/2)$, $t^* \in (-l/2, l/2)$), the distance and minimizing parameters admit the closed-form expressions

$$d_{ij} = \frac{|(\mathbf{t}_i \times \mathbf{t}_j) \cdot (\mathbf{c}_i - \mathbf{c}_j)|}{|\mathbf{t}_i \times \mathbf{t}_j|}, \quad (\text{S6})$$

$$s^* = \frac{(\mathbf{c}_j - \mathbf{c}_i) \cdot \mathbf{t}_i - (\mathbf{t}_i \cdot \mathbf{t}_j) (\mathbf{c}_j - \mathbf{c}_i) \cdot \mathbf{t}_j}{1 - (\mathbf{t}_i \cdot \mathbf{t}_j)^2}, \quad (\text{S7})$$

$$t^* = \frac{(\mathbf{t}_i \cdot \mathbf{t}_j) (\mathbf{c}_j - \mathbf{c}_i) \cdot \mathbf{t}_i - (\mathbf{c}_j - \mathbf{c}_i) \cdot \mathbf{t}_j}{1 - (\mathbf{t}_i \cdot \mathbf{t}_j)^2}. \quad (\text{S8})$$

For the general cases (parallel rods or endpoint contacts), analytic formulas are available and can be evaluated efficiently using the algorithm of [6].

The nearest points and contact normal are

$$\mathbf{x}_i = \mathbf{c}_i + s^* \mathbf{t}_i, \quad \mathbf{x}_j = \mathbf{c}_j + t^* \mathbf{t}_j, \quad \mathbf{n}_{ij} = \frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}. \quad (\text{S9})$$

Differentiating g_{ij} along the system trajectory gives the velocity-level constraint

$$\dot{g}_{ij} = \mathbf{n}_{ij} \cdot (\dot{\mathbf{c}}_i - \dot{\mathbf{c}}_j) + (s^* \mathbf{t}_i \times \mathbf{n}_{ij}) \cdot \boldsymbol{\omega}_i - (t^* \mathbf{t}_j \times \mathbf{n}_{ij}) \cdot \boldsymbol{\omega}_j = \mathbf{J}_{ij}(\mathbf{q}) \mathbf{v}, \quad (\text{S10})$$

where $\mathbf{v}$ stacks all rod twists, and

$$\mathbf{J}_{ij} = [\mathbf{n}_{ij}^T \quad (s^* \mathbf{t}_i \times \mathbf{n}_{ij})^T \quad -\mathbf{n}_{ij}^T \quad -(t^* \mathbf{t}_j \times \mathbf{n}_{ij})^T] \quad (\text{S11})$$

is the contact row of the full constraint Jacobian $\mathbf{J}(\mathbf{q}) = \nabla_{\mathbf{q}} \mathbf{g}(\mathbf{q})$.

The block mass matrix is $\mathbf{M} = \text{diag}(\mathbf{M}_1, \dots, \mathbf{M}_N)$, with

$$\mathbf{M}_i = \begin{bmatrix} m \mathbf{I}_3 & 0 \\ 0 & \mathbf{I}_i \end{bmatrix}, \quad (\text{S12})$$

where m is the rod mass and $\mathbf{I}_i$ is its inertia tensor about the center. Between contact events ($g_{ij} > 0$), the motion is smooth and governed by the Newton–Euler equations at the acceleration level,

$$\mathbf{M}(\mathbf{q}) \dot{\mathbf{v}} + \mathbf{J}(\mathbf{q})^T \mathbf{f} = \mathbf{0}, \quad (\text{S13})$$

where $\mathbf{f} \geq \mathbf{0}$ are contact forces satisfying the complementarity conditions $g_{ij} \geq 0$, $f_{ij} \geq 0$, and $g_{ij} f_{ij} = 0$. This formulation is an acceleration-level (force-based) description of the constrained dynamics.

At contact ($g_{ij} = 0$), however, the acceleration-level formulation may become ill-posed in the presence of friction, leading to indeterminacy or inconsistency (the Painlevé paradox) [8]. To resolve this, a velocity-level (impulse-based) formulation at impact is often adopted:

$$\mathbf{M}(\mathbf{q})(\mathbf{v}^+ - \mathbf{v}^-) = \mathbf{J}(\mathbf{q})^T \boldsymbol{\lambda}, \quad (\text{S14})$$

where $\boldsymbol{\lambda} = \int_{t^-}^{t^+} \mathbf{f} dt \geq \mathbf{0}$ are the contact impulses and $\mathbf{v}^\pm$ are the pre- and post-impact velocities, respectively. The unilateral conditions impose the complementarity relations

$$g_{ij}(\mathbf{q}) \geq 0, \quad \lambda_{ij} \geq 0, \quad g_{ij}(\mathbf{q}) \lambda_{ij} = 0, \quad (\text{S15})$$

enforcing non-penetration and ensuring that impulses act only at active contacts. Tangential impulses are governed by the Coulomb friction law.

S3 Nondimensionalization

We show that the geometry and kinematics of rigid non-overlap depend only on the aspect ratio $\alpha = l/d$, while additional physics introduces further dimensionless groups.

S3.1 Geometric nondimensionalization

Using rod length l as the reference length, define $\tilde{\mathbf{c}}_i = \mathbf{c}_i/l$, $\tilde{s} = s/l$, so that $\tilde{\mathbf{r}}_i(\tilde{s}) = \tilde{\mathbf{c}}_i + \tilde{s} \mathbf{t}_i$ with $\tilde{s} \in [-1/2, 1/2]$. The nondimensional gap

$$\tilde{g}_{ij} = \frac{g_{ij}}{l} = \tilde{d}_{ij} - \alpha^{-1} \quad (\text{S16})$$

depends only on the dimensionless geometry, hence only on α .

S3.2 Nondimensionalization of the velocity-level constraint

Let T be a characteristic time scale and define $\tilde{\mathbf{v}}_i = T\dot{\mathbf{c}}_i/l$ and $\tilde{\boldsymbol{\omega}}_i = T\boldsymbol{\omega}_i$. Substituting into $\dot{g}_{ij} = \mathbf{J}_{ij}\mathbf{v}$ gives

$$\frac{T}{l} \dot{g}_{ij} = \tilde{\mathbf{J}}_{ij}(\tilde{\mathbf{q}}; \alpha) \tilde{\mathbf{v}}, \quad (\text{S17})$$

where $\tilde{\mathbf{J}}$ depends only on the dimensionless configuration $\tilde{\mathbf{q}}$ and α .

S3.3 Scaling of mass and inertia

Consider a solid cylinder of constant mass m . The transverse and axial moments of inertia are

$$I_{\perp} = \frac{1}{12}m(3r^2 + l^2), \quad I_{\parallel} = \frac{1}{2}mr^2, \quad (\text{S18})$$

where $r = d/2$. In the slender limit $r \ll l$,

$$I_{\perp} \approx \frac{1}{12}ml^2 = O(ml^2), \quad I_{\parallel} \approx \frac{1}{8}md^2 = O(ml^2\alpha^{-2}). \quad (\text{S19})$$

The dimensionless inertia tensor $\hat{\mathbf{I}}_i(\alpha) = \mathbf{I}_i/(ml^2)$ therefore depends only on α . This conclusion remains valid if, instead, one assumes constant density, in which case $m \sim \rho l d^2$ varies with α . However, the overall mass scale cancels in the relative velocity dynamics (see End Matter), which are governed by ratios of Delassus components (e.g., W_{nm}/W_{tt}), and are therefore independent of m .

S3.4 Dimensionless equations of motion

Writing $\mathbf{M}_i = ml^2\hat{\mathbf{M}}_i(\alpha)$ and nondimensionalizing Eq. (S14) with scales l , m , T gives

$$\hat{\mathbf{M}}(\alpha) \frac{d\tilde{\mathbf{v}}}{d\tilde{t}} = \tilde{\mathbf{J}}(\tilde{\mathbf{q}}; \alpha)^T \tilde{\boldsymbol{\lambda}}, \quad (\text{S20})$$

subject to $\tilde{g}(\tilde{\mathbf{q}}; \alpha) \geq 0$. For purely inertial rigid contact, all rod-shape dependence enters through α . Additional physics introduces further dimensionless groups: gravity adds $\Gamma_g = gT^2/l$; translational and rotational drag add $\Gamma_t = \zeta_t T/m$ and $\Gamma_r = \zeta_r T/(ml^2)$; soft steric repulsion with stiffness k adds $\Gamma_k = kdT^2/(ml) = (kT^2/m)\alpha^{-1}$. The coefficient of restitution e and friction coefficient μ are already dimensionless. A rigid elastic frictional system is therefore fully characterized by $(\alpha, \mu, \varepsilon)$ unless any external forcing groups are added to the dynamics.

S3.5 Numerical implementation of rigid body dynamics under contact and friction constraints

The packings are generated using our optimization protocols and are characterized by their gap-to-length ratios ($\tilde{g}_t$ and $\tilde{g}_r$) and friction coefficient μ . Each rod is initialized with random translational and rotational perturbations of comparable kinetic scale. Specifically, the translational and rotational initial velocities $\dot{\mathbf{c}}_0$ and $\boldsymbol{\omega}_0$ were sampled from the Gaussian distributions, $\mathcal{N}(0, v_0^2)$ and $\mathcal{N}(0, (2v_0/l)^2)$. The azimuthal angular velocity is then removed after the sampling.

The rigid-body dynamics are advanced using a semi-implicit Euler time-stepping scheme with non-penetration and dry-friction constraints. Upon contact events ($g < 0$), contact impulses and the corresponding velocity updates are obtained by solving a cone complementarity problem via a projected Gauss-Seidel iteration, following the matrix-free formulation implemented in Project Chrono [4, 9, 10]. No rolling friction torque is applied in the solver. The numerical implementation of the rigid-body dynamics for rods is available in a GitHub repository: <https://github.com/yeonsu-jung/rod-dynamics-3d>.

Parameter	Symbol	Default value
Rod length	l	1.0
Rod diameter	d	from input
Density	ρ	1000.0
Time step	Δt	0.001
Friction coefficient	μ	from input
PGS velocity iterations	–	200
PGS Baumgarte factor	β	0.0
PGS constraint force mixing	–	0.05
PGS relaxation parameter	ω	1.0

Table S1: Default simulation parameters for the bundled default scene in rod-dynamics-3d. The rod mass is computed as $m = \rho\pi r^2 l$, where $r = d/2$.

Gravity and damping are not included in this simulation.

S3.6 Time scales

We set the length of rod and the characteristic translational and rotational speed as free parameters, or $l = 1$ and $v_0 = 0.1$ and $\omega_0 = v_0/(l/2) = 0.2$. Time scale can be determined by combining these two length and velocity scales. An easy choice is l/v_0 . It simply means how long it would take for a rod to travel by its length with constant speed v_0 or rotate following the arc of length l with ω_0 . However, this does not scale with how fast the entanglement decays with v_0 . Thus, we use untanglement time as follows: how long would it take for ACN to become half if one of the rods with length 1 is moving away from the

other one which was originally intersecting and transverse. From Eq. S25, it is the solution of $x^2(8 + 16x^2) = 1$ ($x = \pm 0.32$), meaning if the distance is length multiplied by 0.32, then the ACN will be 0.25 which is the half of the original value, 0.5.

S4 No finite rod packing is rigid

We show that a packing of finitely many rods in $\mathbb{R}^3$ cannot be infinitesimally rigid. For two skew rods i and j in contact at interior points ($s^*, t^* \in (-l/2, l/2)$), the rate of change of the gap is

$$\dot{g}_{ij} = \mathbf{n}_{ij} \cdot (\dot{\mathbf{c}}_i + s^* \dot{\mathbf{t}}_i - \dot{\mathbf{c}}_j - t^* \dot{\mathbf{t}}_j). \quad (\text{S21})$$

This vanishes not only for the six global rigid-body modes of the packing, but also for motions that are *soft* with respect to contact ij :

- sliding along the rod axis: $\mathbf{n}_{ij} \cdot \dot{\mathbf{c}}_i = 0$,
- spinning around the contact normal: $\mathbf{n}_{ij} \cdot \dot{\mathbf{t}}_i = 0$,
- compensating relative translation and rotation: $\dot{\mathbf{c}}_i + s^* \dot{\mathbf{t}}_i = \dot{\mathbf{c}}_j + t^* \dot{\mathbf{t}}_j$.

A packing cannot be infinitesimally rigid because the outermost rod—identified from the convex hull of all rod endpoints—can always be pulled outward along a soft sliding direction. Formally, $\dot{g}_{ij} = 0$ for all contacts does not imply $\dot{\mathbf{q}} = 0$, since $\mathbf{n}_{mj} \cdot \dot{\mathbf{t}}_m = 0$ for any rod m on the convex hull. Because any finite set of points has a convex hull, this statement holds for any finite packing. It remains true even in the presence of friction: pulling the outermost rod outward with infinite precision drives the normal force, and hence the tangential friction force, to zero.

S5 Average crossing number

The average crossing number (ACN) for a pair of rods i and j is

$$\text{ACN}_{ij} = \frac{1}{4\pi} \iint_{[-l/2, l/2]^2} \frac{|(\mathbf{t}_i \times \mathbf{t}_j) \cdot (\mathbf{r}_i - \mathbf{r}_j)|}{\|\mathbf{r}_i - \mathbf{r}_j\|^3} ds dt, \quad (\text{S22})$$

where $\mathbf{r}_i = \mathbf{r}_i(s)$ and $\mathbf{r}_j = \mathbf{r}_j(t)$. This measures the average solid angle of directions for which the two rods intersect in projection, divided by 4π [2]. The integral can be evaluated analytically [5]:

$$\begin{aligned} \text{ACN}_{ij} = & F(a_1 + l, a_2 + l) - F(a_1 + l, a_2) \\ & - F(a_1, a_2 + l) + F(a_1, a_2), \end{aligned} \quad (\text{S23})$$

where

$$F(x, y; d_{ij}, \theta) = \frac{1}{4\pi} \arctan \left(\frac{(x/d_{ij})(y/d_{ij}) \sin \theta + \cos \theta}{\sqrt{(x/d_{ij})^2 + (y/d_{ij})^2 - 2(x/d_{ij})(y/d_{ij}) \cos \theta + 1}} \right), \quad (\text{S24})$$

with $\theta = \arccos(\mathbf{t}_i \cdot \mathbf{t}_j)$ the angle between rods and a_1, a_2 the signed skewness parameters (offsets of s^*, t^* from the midpoints). In the symmetric case $\theta = \pi/2$ and $a_1 = a_2 = 0$, this simplifies to

$$\text{ACN}_{ij} = \frac{1}{\pi} \arctan \left(\frac{1}{d_{ij} \sqrt{8 + 16d_{ij}^2}} \right), \quad (\text{S25})$$

which increases monotonically toward $1/2$ as $d_{ij} \rightarrow 0$ (see Fig. S1), consistent with the intuition that two perpendicular rods crossing at their midpoints are maximally entangled.

The ACN is sensitive to four degrees of freedom beyond overall rigid-body motion: the normal separation d_{ij} , the crossing angle θ , and the contact-anchor positions (s^*, t^*) . Equation (S25) shows that to maximize ACN_{ij} one should minimize d_{ij} (contact), maximize θ (prefer perpendicular orientations), and center the contact near the rod midpoints ($a_1, a_2 \approx 0$). For $N > 3$ rods, not all pairs can simultaneously satisfy $\theta_{ij} = \pi/2$, and the problem of finding the optimal N -rod configuration reduces to packing lines through a point in $\mathbb{R}^3$, related to the Grassmannian problem [3].

In Fig. S2 we show that the normalized entanglement of maximally entangled packings approaches its maximum value for $\alpha \gg 100$, regardless of N . Decreasing α (increasing d at fixed $l = 1$) reduces $\tilde{e}$ because the non-overlap constraint prevents full lateral contact.

Numerically, the optimization variable is the collection of all rod coordinates $\mathbf{q}$. For each unordered pair of rods (i, j) , the code evaluates the straight-segment average crossing number, which for line segments coincides here with the magnitude of the linking number. Using the formula for polygonal curves[7], one forms the four normalized face normals (n_1, n_2, n_3, n_4) from the endpoint difference vectors and computes

$$s_{ij} = \arcsin(n_1 \cdot n_2) + \arcsin(n_2 \cdot n_3) + \arcsin(n_3 \cdot n_4) + \arcsin(n_4 \cdot n_1),$$

so that the pair contribution is taken as

$$\text{ACN}_{ij} = -\frac{1}{4\pi} \sqrt{s_{ij}^2 + \epsilon},$$

with a small ϵ included only to smooth the absolute value at zero. The total objective is therefore

$$f(q) = -\sum_{i < j} \text{ACN}_{ij} = -e(\mathbf{q}),$$

so minimizing $f(q)$ is equivalent to maximizing the total pairwise average crossing number, or equivalently the average over pairs after dividing by $N(N-1)/2$. Gradients $\nabla_{\mathbf{q}} f(\mathbf{q})$

are obtained by automatic differentiation in JAX, and the rod configuration is updated with the FIRE optimizer[1], which iteratively moves the centroids and orientations in the direction that increases the total pairwise linking while preserving the straight-rod parametrization. The numerical implementation of the average crossing number and the associated optimization is available in a GitHub repository: <https://github.com/yeonsu-jung/entanglement-optimization>. The hyper-parameters to run FIRE optimization are:

```
_NDELAY = 10
_FINC   = 1.1
_FDEC   = 0.5
_FA     = 0.99
_ALPHA0 = 0.1
dt      = 1.e-5
dt_max  = 1.0
```

S6 Contact criterion for static packing

In fully relaxed packings obtained using the repulsive potential, inter-rod distances are strictly greater than d up to a numerical tolerance of $\sim 10^{-9}$, set by the effective step size of the relaxation dynamics. We therefore define a contact as a pair of rods whose separation is smaller than $1.01d$. The optimized packings exhibit $Z \approx 4$, a reduced coordination number relative to $Z \approx 10$ in random packings, which we attribute to the strong spatial-orientational correlations induced by entanglement maximization.

S7 Free volumes in configuration space

The configuration space of a rigid rod is the Lie group $\text{SE}(3)$. Fixing all other rods $j \neq k$, the collision-free region accessible to rod k is the connected component

$$D_k = C_{\mathbf{q}_k} \{ \mathbf{q} \in \text{SE}(3) : g_{kj}(\mathbf{q}) \geq 0, \forall j \neq k \}, \quad (\text{S26})$$

where $C_{\mathbf{q}_k}$ denotes the component containing $\mathbf{q}_k$. The free mobility volume is

$$F_k = \int_{D_k} d\mu(\mathbf{q}), \quad (\text{S27})$$

with $d\mu$ the Haar measure on $\text{SE}(3)$.

For a unit direction $\mathbf{u} \in \mathbb{S}^{n-1}$ in the tangent space $T_{\mathbf{q}_k}\text{SE}(3)$ (with n the effective dimension), the directional free path is

$$r_k(\mathbf{u}) = \sup\{r \geq 0 : \mathbf{q}_k + r\mathbf{u} \in D_k\}. \quad (\text{S28})$$

Assuming D_k is locally star-shaped,

$$F_k \approx \frac{1}{n} \int_{\mathbb{S}^{n-1}} r_k(\mathbf{u})^n d\Omega(\mathbf{u}). \quad (\text{S29})$$

The ACN between rods k and j , in the limit $d \rightarrow 0$, measures the solid angle of translational directions along which the rods would intersect in projection. Since closer, more transverse rod pairs obstruct a larger fraction of directions, ACN serves as a robust proxy for confinement: higher ACN implies tighter self-caging.

One might expect that decreasing d at fixed l would tighten the packing and reduce the accessible free volumes, with scalings $\sqrt{A^*} \sim d$ and $\sqrt{\Omega^*} \sim d/l \sim 1/\alpha$. Our results suggest otherwise: at large aspect ratios ($\alpha \sim 10^3$) we observe $\sqrt{A^*} \approx 10^{-2} \gg d \sim 10^{-3}$ and $\Omega^* \approx 10^{-2} \gg d/l \sim 10^{-3}$. The free volumes do not collapse to zero as $d \rightarrow 0$; instead, confinement in the slender-rod limit is governed by the geometric arrangement of neighboring rods, which imposes a finite bound on accessible motions independently of rod thickness.

S8 Numerical free-volume calculation

Each rod is represented as an identical spherocylinder with endpoints $\mathbf{p}_1$ and $\mathbf{p}_2$ and diameter d where $\|\mathbf{p}_1 - \mathbf{p}_2\| = l = 1$. For a tagged rod k , we measure translational and rotational free volume while all neighboring rods are held fixed. Two objects are taken to overlap when the minimum distance between the corresponding line segments is smaller than d .

To reduce the cost of collision tests, we first construct an axis-aligned bounding box around the tagged rod and expand it by a conservative displacement range that covers both translations and rotations. Only rods whose own bounding boxes overlap this expanded box are retained as candidate neighbors. All subsequent collision checks are restricted to this list.

For translational free volume, we build an orthonormal frame $\{\mathbf{d}_1, \mathbf{d}_2, \mathbf{t}\}$, where $\mathbf{t}$ is the rod axis, and sample in-plane directions $\psi \in [0, 2\pi)$. For each ψ , the rod is displaced by $\Delta(\cos \psi \mathbf{d}_1 + \sin \psi \mathbf{d}_2)$, and we determine the first-contact distance $d_k(\psi)$ by a coarse radial scan followed by bisection. Here bisection means repeatedly halving the interval between the largest non-overlapping trial displacement and the smallest overlapping one until the contact distance is resolved to the desired tolerance. The translational free area is then evaluated as

$$A_k = \int_0^{2\pi} \frac{1}{2} d_k^2(\psi) d\psi, \quad (\text{S30})$$

with the minimum translational clearance given by $\min_\psi d_k(\psi)$.

For rotational free volume, we rotate the tagged rod about its center while keeping its length fixed. Writing the trial orientation as

$$\mathbf{t}'(\theta, \phi) = \sin \theta \cos \phi \mathbf{d}_1 + \sin \theta \sin \phi \mathbf{d}_2 + \cos \theta \mathbf{t}, \quad (\text{S31})$$

Option	Default	Description
<code>--diameter <d></code>	from header	Rod diameter
<code>--radius <r></code>	from header	Rod radius (= diameter/2)
<code>--samples <n></code>	360	Angular sample count
<code>--bisection-steps <n></code>	16	Bisection refinement steps
<code>--theta-coarse <n></code>	48	Coarse θ scan steps
<code>--measure-single-rod <i></code>	off	Detailed profile for rod i
<code>--output <file></code>	stdout	Output file path
<code>--threads <n></code>	auto	OpenMP thread count
<code>-v, --verbose</code>	off	Progress to stderr

Table S2: Command-line options for **rod-free-volume** binary.

we sample azimuthal angles $\phi \in [0, 2\pi)$ and, for each ϕ , determine the first-contact polar angle $\theta_k(\phi)$ by the same coarse-scan-plus-bisection procedure. The rotational free volume is reported as the accessible solid angle,

$$\Omega_k = \int_0^{2\pi} [1 - \cos \theta_k(\phi)] d\phi, \quad (\text{S32})$$

and the minimum rotational clearance is $\min_\phi \theta_k(\phi)$. In practice, both integrals are evaluated by uniform angular quadrature, and rods are processed independently, allowing straightforward shared-memory parallelization. The numerical implementation of the free volume calculation is available in a GitHub repository: <https://github.com/yeonsu-jung/rod-free-volume>.

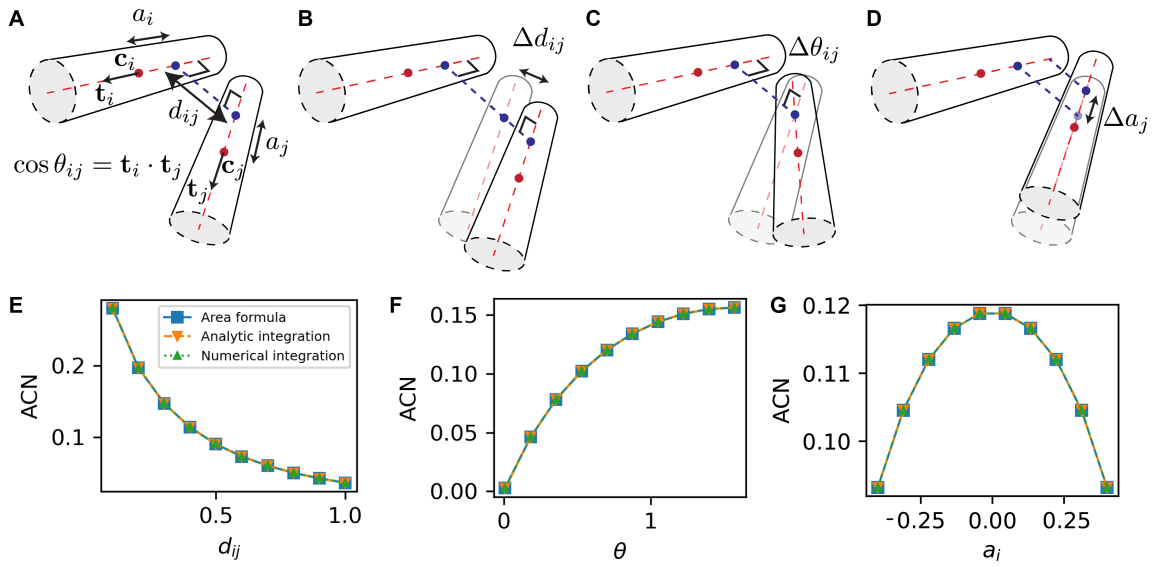


Figure S1: **Definition and behavior of the average crossing number (ACN).** (A–D) Geometry of two skew rods, illustrating variation in distance d_{ij} , angle θ , and skewness parameters a_i , a_j . (E–G) ACN as a function of d_{ij} (E), θ (F), and a_i (G).

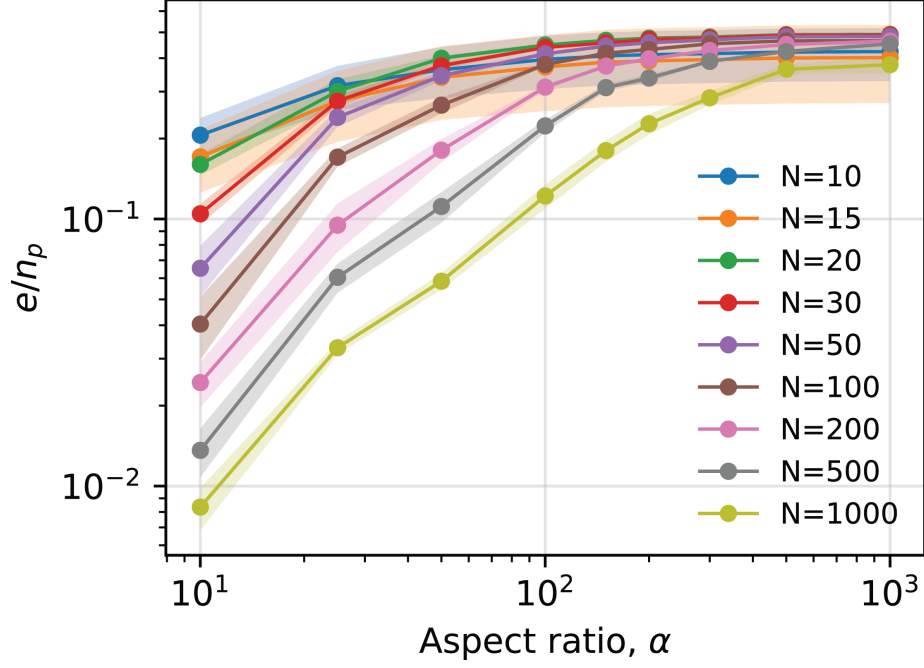


Figure S2: Normalized entanglement $\tilde{e}$ of maximally entangled packings for multiple values of N as a function of aspect ratio α . The packings were generated using the Python scripts `entangle.py` and `relax.py` from the repository <https://github.com/yeonsu-jung/entanglement-optimization>. The parameters used for this plot are the default values listed in Sec. S5 and in the repository's `README.md`.

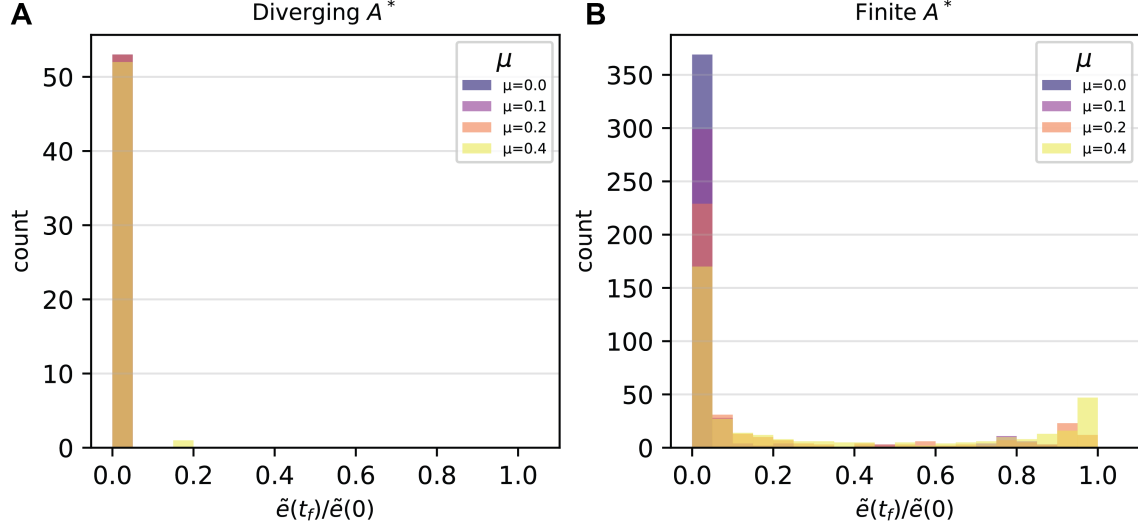


Figure S3: **Softness of non-self-caged packings.** Histograms of the entanglement retention ratio $\tilde{e}(t_f)/\tilde{e}(0)$ for packings with diverging (A) and finite (B) values of A^* . The entangled and relaxed packings, $\mathbf{q}^{**}$, were generated using `entangle.py` and `relax.py` from <https://github.com/yeonsu-jung/entanglement-optimization>. The maximum translational free volume per rod, A^* , was computed using the `rod-free-volume` binary compiled from <https://github.com/yeonsu-jung/rod-free-volume>. The retention ratio was then measured from rigid-body simulations over multiple friction coefficients and initial-velocity realizations, using the `rigidbody-viewer-3d` binary compiled from <https://github.com/yeonsu-jung/rod-dynamics-3d>. The parameters used for this plot are the default values listed in Sec. S5 and Tables S2 and S1.

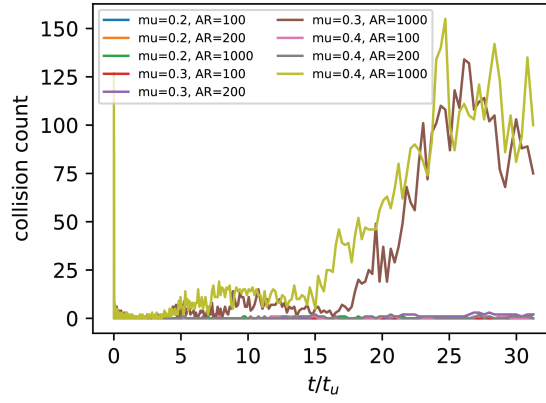


Figure S4: Emergence of sticking dry friction at later stages based on the hard contact model we used [9, 10] in <https://github.com/yeonsu-jung/rod-dynamics-3d>. Initially, the number of collision per time step quickly drops by slowed contact velocities. Later, the friction force becomes within the Coulomb cone and serves as persisting contact forces to balance the centrifugal force associated with the global rotation.